



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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Lecture Notes-2.5

22GE004–BASICS OF ELECTRONICS ENGINEERING

UNIT II & SIGNAL CONDITIONING USING DIODE

TOPIC:SIGNAL CLIPPING

Clipper Circuit

A Clipping Circuit plays a vital role in removing unwanted portions of the signal. In our daily life, we used to remove the unwanted portion of nails by using a nail cutter. This process of removing unwanted parts is known as clipping and the circuit is called a Clipping circuit. In a similar way, whenever we want to remove the unwanted portion (noise) from the signal that time we will use the electronic circuit accordingly to clip the noise.

Based on the requirement, we will use the respective clipping circuit. p-n junction diodes are having a lot of applications. Clippers are one of those applications.

What is a Clipping Circuit?

An electronic circuit is said to be a clipping circuit if it removes part of the input signal based on the requirement. It is also known as a clipper, in short. A wave shaping circuit is said to be a clipper if it performs one of the following three operations.

- Clips portion or entire positive half part of the input signal
- Clips portion or entire negative half part of the input signal
- Clips portions of both positive and negative half parts of the input signal

Types of Clippers in Clipping Circuit

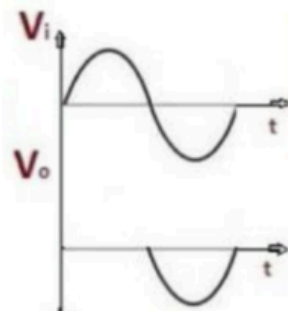
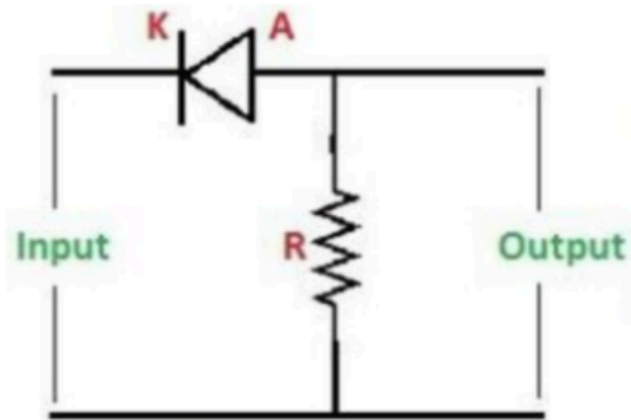
In this article, we are focusing on one of the important applications of Diodes. i.e., clippers. Here, we are assuming all the diodes that we use in the circuit diagrams are ideal. Clippers can be classified based on the portion of the signal that is being clipped and placing of Diode in the circuit. Now, let us discuss the following types of clippers one by one.

- Series Clippers
- Shunt Clippers

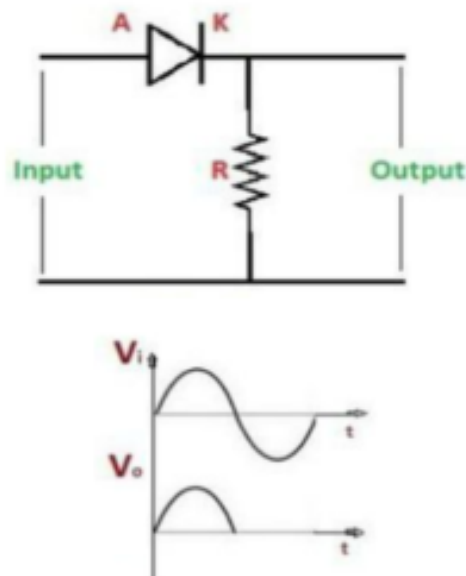
Series Clippers in Clipping Circuit

A clipper is said to be a series clipper if the diode and resistor are connected in series with the input AC (voltage) signal and the output is taken across the resistor. Based on the position of the diode, we can clip either the positive half part or the negative half part of the input signal. Accordingly, we are having the following two types of series clippers.

- **Positive Series Clipper:** A series clipper is said to be a positive series clipper if it clips the positive half part of the input signal. This circuit diagram and the corresponding input-output waveforms are shown in the following figures. There is no output or zero output during the positive half cycle of the input signal since the diode is reverse biased. Output is the same as that of input during the negative half cycle of the input signal since the diode is forward-biased.



- Negative Series Clipper:** A series clipper is said to be a negative series clipper if it clips the negative half part of the input signal. This circuit diagram and the corresponding input-output waveforms are shown in the following figures. There is no output or zero output during the negative half cycle of the input signal since the diode is reverse biased. Output is the same as that of input during the positive half cycle of the input signal since the diode is forward-biased.

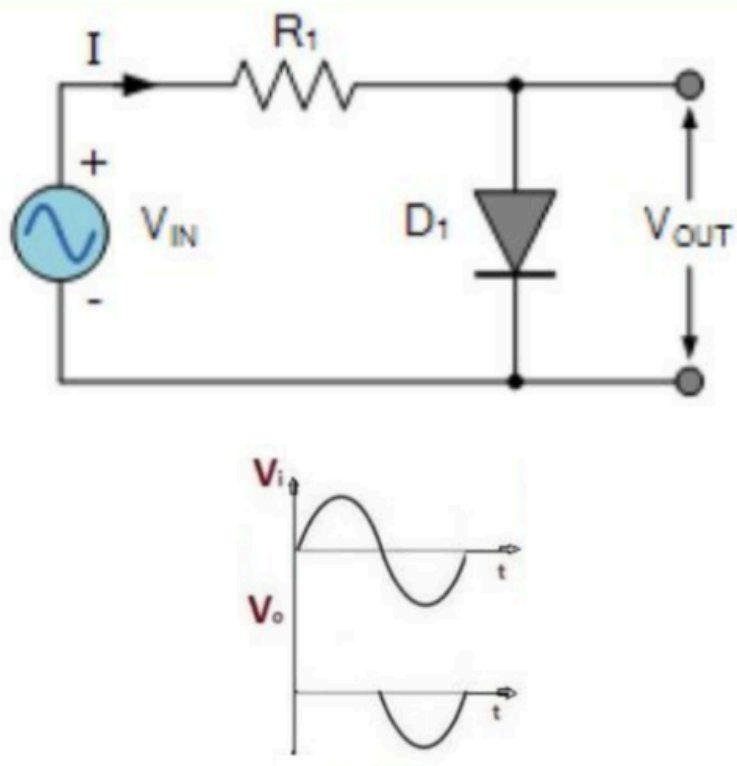


Shunt Clippers in Clipping circuit

A clipper is said to be a shunt clipper if the resistor and diode are connected in series with the input AC (voltage) signal and the output is taken across the diode. Based on the position of the diode, we can clip either the positive half part or the negative half part of the input signal. Accordingly, we are having the following two types of shunt clippers.

- Positive Shunt Clipper:** A shunt clipper is said to be a positive shunt clipper if it clips the positive half part of the input signal. This circuit diagram and the

corresponding input-output waveforms are shown in the following figures. There is no output or zero output during the positive half cycle of the input signal since the diode is forward-biased. Output is the same as that of input during the negative half cycle of the input signal since the diode is reverse biased.



- Negative Shunt Clipper:** A shunt clipper is said to be a negative shunt clipper if it clips the negative half part of the input signal. This circuit diagram and the corresponding input-output waveforms are shown in the following figures. There is no output or zero output during the negative half cycle of the input signal since the diode is forward-biased. Output is the same as that of input during the positive half cycle of the input signal since the diode is reverse

biased.

